Shubh Sahu

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Education

Indian Institute of Technology, Mandi — Himachal Pradesh, India
Bachelor of Technology in Computer Science Engineering; CGPA: 8.77/10 — *Aug 2023 – Present*

Experience

Google — May 2026 – Jul 2026
Software Engineering Intern — *Bengaluru, India*

- Architected Unity Package Manager (UPM) compatibility for the Google Play Games plugin with automated dependency resolution, cutting developer setup time from 5–10 minutes to **under 1 minute**
- Designed a unified monorepo architecture with automated Gradle build tasks to distribute UPM and legacy packages, reducing the committed version-control footprint by **77%** (4.5 MB to under 1 MB)
- Migrated the Play Games network stack from legacy gRPC wrappers to standard xRPC channels using a centralized connection factory, enabling pooled connections over GMS Core's shared Cronet engine and eliminating handshake latency

IIT Mandi — Jan 2026 – May 2026
Research and Development Intern — *Himachal Pradesh, India*

- Built an AI-powered PyQt5 desktop application automating urban seismic risk assessment with PyTorch and TensorFlow, detecting buildings from street-level imagery and classifying them into **24 structural categories**
- Engineered a geospatial risk engine computing earthquake probability and structural financial-loss ratios, visualizing metrics on interactive color-coded Folium maps to support city officials' safety planning
- Architected an automated pipeline to acquire Google Street View imagery via API and generate 3D spatial-twin models, integrating drone photogrammetry (NodeODM) to cut manual structural-analysis effort

Projects

Distributed NoSQL Key-Value Database — GitHub
Java 21, Spring Boot, gRPC/Protobuf, Raft, Docker Compose, React/TypeScript — *Jul 2026 – Present*

- Engineered a distributed key-value store in Java with a custom LSM-Tree engine — WAL, ConcurrentSkipListMap memtable, **8MB**-threshold flush to immutable SSTables — using Bloom Filters and an LRU cache to minimize disk I/O
- Designed a stateless coordinator routing traffic across 3 storage nodes via consistent hashing (MD5, **150 virtual nodes/node**) to prevent hotspots, using gRPC/Protobuf for high-performance internal node communication
- Implemented the Raft consensus algorithm from scratch — leader election, RequestVote/AppendEntries RPCs, randomized **150–300ms** election timeout — for automatic failover and quorum-based log replication

TongueLingua — GitHub
Jetpack Compose, Kotlin, FastAPI, OpenCV, SAM, Gemini API — *May 2025*

- Developed an Android diagnostics app for real-time tongue health analysis in Jetpack Compose and Kotlin, cutting manual inspection time by **85%** using OpenCV and Segment Anything Model (SAM) pipelines
- Built a modular FastAPI backend with clean API design and database integration to handle diagnostic image processing and result storage
- Integrated a Gemini-API-powered chat interface for natural-language health queries, lifting user engagement by **40%**

Technical Skills

Languages: Python, JavaScript, Java, C++, Kotlin, Shell, SQL, HTML/CSS

Backend: FastAPI, Django, gRPC, MongoDB, Prisma, Firebase

Frontend/Mobile: React.js, Tailwind CSS, Jetpack Compose, Unity

AI/ML: Transformers, RAG Pipelines, GANs, Diffusion Models, VAEs, CNNs

Relevant Coursework

Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Deep Learning & Generative AI, Natural Language Processing, Statistical Learning

Achievements & Leadership

- 1st Place, KrackHack 2025 Hackathon, IIT Mandi | 3rd Place, Aptos Blockchain Hackathon, IIT Mandi
- 1st Place (Problem Statement), Deep Learning Hackathon, IIT Mandi | CodeChef 1750+, Codeforces 1560 rating
- Teaching Assistant for Deep Learning, IIT Mandi – mentored 200+ students